Knowledge-Guided and Machine-Learning-Assisted Synthesis for Series-Fed Microstrip Antenna Arrays Using Base Element Modeling

Weiqi Chen, *Graduate Student Member, IEEE*, Qi Wu, *Member, IEEE*, Jiahao Wei, Chen Yu, *Member, IEEE*, Haiming Wang, *Member, IEEE*, and Wei Hong, *Fellow, IEEE*

***Abstract*— The complexity of the series-fed microstrip antenna array (SFMAA) synthesis problem increases rapidly with increasing element number. When dealing with complex practical SFMAA synthesis problems, conventional ideal antenna-based and electromagnetic (EM) full-wave simulation-based methods are trapped in performance degradation and time-consuming iteratively "cut-and-try" processes. Machine-learning-assisted (MLA) methods face the "curse of dimensionality," leading to significantly increased training and prediction times and reduced prediction performance. Here, base element (BE) modeling is used to develop a knowledge-guided and MLA synthesis method for SFMAAs. First, the similarity of BEs with different locations and dimensions is explored using knowledge-guided classification. Then, a cosine-domain learning method with low computational costs is introduced. Next, array synthesis, microwave network cascades, and MLA optimization approaches are used to design SFMAAs. This hybrid knowledge-guided and data-driven method greatly facilitates the synthesis process, achieving low training and prediction times and high accuracy. Furthermore, synthesis applications with three types of SFMAAs and various practical design goals are utilized to verify the effectiveness of the proposed method. Finally, the designed antenna prototypes are fabricated, and the measured results show excellent agreement with the simulation results.**

***Index Terms*— Base element modeling (BEM), discrete cosine transform (DCT), knowledge-guided, machine learning (ML), series-fed microstrip antenna array (SFMAA).**

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Weiqi Chen and Chen Yu are with the State Key Laboratory of Millimeter Waves, School of Information Science and Engineering, Southeast University, Nanjing 210096, China (e-mail: wq_chen@seu.edu.cn; chenyu@seu.edu.cn).

Qi Wu, Haiming Wang, and Wei Hong are with the State Key Laboratory of Millimeter Waves, School of Information Science and Engineering, Southeast University, Nanjing 210096, China, and also with the Pervasive Communication Research Center, Purple Mountain Laboratories, Nanjing 211111, China (e-mail: qiwu@seu.edu.cn; hmwang@seu.edu.cn; weihong@seu.edu.cn).

Jiahao Wei was with the State Key Laboratory of Millimeter Waves, School of Information Science and Engineering, Southeast University, Nanjing 210096, China. He is now with COT Design Department, Hisilicon Technologies Company Ltd., Shenzhen 518129, China (e-mail: weijiahao7@huawei.com).

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I. Introduction

THE series-fed microstrip antenna array (SFMAA) is one of the most ubiquitous array structures. It has been widely investigated and applied in wireless communication, microwave sensors, and radar systems due to its advantages of a low profile, low cost, simplified feeding mechanism and reduced feed line loss [1], [2]. To meet the requirements of specific application scenarios, such as narrow beams, low sidelobe levels (SLLs), and cosecant squared patterns, various SFMAA-based pattern synthesis methods have been proposed [3], [4], [5], [6], [7], [8], [9].

The transmission line model represents patch elements as two radiating narrow parallel slots; this is the easiest method to analyze SFMAAs and elucidates the physical results [1], [10]. In addition, circuit representations of patches were established, and the equivalent parameters were calculated experimentally [4]. In [3], the radiative conductance, voltage, and current required to excite the desired pattern were calculated using the equivalent circuit and transmission line equations. A set of canonical coefficients obtained through full-wave simulations for SFMAA elements was defined in [6]. After the surface current density at the center of the elements was obtained, the radiation pattern of the array was determined.

Patch width tapering and linewidth tapering are commonly used approaches to meet the specific design requirements of low SLLs [5], [7]. The radiation element can be considered a two-port microwave network that cascades into an SFMAA [7], and the relationship between the linewidth and array element directivity can be obtained through electromagnetic (EM) full-wave simulations. The radiation pattern of the patch was regarded as two point sources in [8], and a formula for the radiation pattern of the array was established. They used the differential evolution algorithm (DEA) to design a low SLL and narrow-beam array with unequal interelement spacing and nonuniform amplitude. The DEA and full-wave simulations were applied to design a low SLL and narrow-beam automotive radar array at 77 GHz by optimizing the width of the elements and the interelement spacing [9].

In contrast to conventional gradient-based algorithms, metaheuristic algorithms such as genetic algorithms (GAs) [11], DEAs, and particle swarm optimization (PSO) algorithms

[12] can effectively avoid getting trapped in local optima. Unfortunately, these algorithms typically require hundreds of function calls to converge [13]. For low-complexity functions, the computational cost is primarily due to the optimization process, while for EM full-wave simulation-based synthesis problems, the computational cost is typically due to the full-wave simulations [14]. To reduce the computational costs, machine-learning (ML) methods have attracted considerable attention and have been extensively studied in EM device design [15], [16], [17], [18], [19], [20]. Existing ML-assisted optimization (MLAO) algorithms can be divided into offline model-based methods [21] and online model-based methods [15]. Offline data-driven optimization methods generally require more initial samples, and optimization is performed after training to obtain the surrogate models, which will not be updated again. The online model updates the surrogate model through new sample points during the optimization process to improve the prediction accuracy. Both methods basically involve the following steps: sampling, training surrogate models with ML algorithms, optimizing surrogate models with global or local optimization algorithms, and verifying the optimization results with EM full-wave simulations. Most practical EM-related problems that can be addressed by conventional MLAO algorithms are small-scale problems.

As the dimensionality of the problem increases, the difficulty of sampling, training, and optimization increases considerably [14]. On the one hand, the number of samples required for ML model training increases substantially due to the "curse of dimensionality." Sampling increases the computational cost of array full-wave simulations. On the other hand, the training and prediction times of ML models cannot be neglected as the amount of data increases [22], [23]. For surrogate model training and optimization, dimensionality reduction techniques are typically used to address medium-scale problems with 20–50 variables [22]. The self-adaptive Gaussian process (GP) surrogate model and hybrid surrogate model have been proposed to optimize practical antennas with high-dimensional and multiple specifications [23]. A Bayesian neural network-assisted global optimization combined with a self-adaptive lower confidence bound strategy is proposed to enhance the convergence speed and reduce the training cost of ML [24].

However, the considerable sampling time is not desirable for pure data-driven strategies. Online updating surrogate models using prescreening strategies can, to some degree, mitigate the "curse of dimensionality" [15], [22], [23], [25]. To reduce the computational cost of the design process, coarse-discretization EM models of antenna structures were applied in [16]. Recently, knowledge-guided active base element modeling (ABEM) of antenna arrays was proposed [26], [27]. Many active base element patterns (ABEPs) can be obtained with only one array simulation, and only the effects of adjacent elements and the platform on the ABEP are considered during the training process [26]. Compared with conventional ML-assisted (MLA) array optimization methods, this method improves the training accuracy while reducing the number of required samples and training time. A planar virtual subarray approximation method was also proposed, and the training cost of the 3-D ABEP method was significantly reduced [27].

In contrast to antenna arrays with individually fed elements, the ABEPs of SFMAAs are difficult to obtain in array environments. Moreover, ABEM-based methods can be applied with elements with the same structures and dimensions [26], [27] and are thus less applicable to SFMAA designs. In this work, a knowledge-guided MLA-SFMAA synthesis method based on base element (BE) modeling is proposed. First, an SFMAA is characterized by three types of BEs using prior knowledge. To further alleviate the computational burden, the discrete cosine transform (DCT) is introduced to compress the BE pattern (BEP) data. Next, the *S*-parameters of the array are approximated by a cascade of the *S*-parameters of BEs. The radiation pattern of the array is approximated from the synthesis of the patterns of BEs.

The current of each antenna element in the array depends not only on the excitation, but also on the effect of mutual coupling (MC) with all other neighboring antennas. The interelement coupling in the SFMAAs can be partitioned into conductive and radiative ones. The effect of MC must be considered in the model to accurately predict the radiation and impedance characteristics of the array [28]. Array synthesis methods based on active element patterns (AEPs) are effective for considering the effect of MC [26], [27]. However, as previously mentioned, AEPs are difficult to acquire for SFMAAs. Thus, in this work, the conductive coupling is impliedly taken into account in the network cascade. The interelement conductive coupling is considered the main factor causing performance degradation, and the interelement radiative coupling is supposed to be ignored. In addition, low-fidelity (LF) verification and local optimization strategies are introduced in the optimization process to compensate for the effects of ignoring radiative coupling. The final optimization results show that the array performance degradation during the synthesis process is acceptable. The main contributions of this work can be summarized as follows.

1) The method of cascading characterization SFMAA of BEs greatly reduces computational costs while ensuring high prediction accuracy. The hybrid knowledge-guided and data-driven BE modeling (BEM) method is more efficient than the purely data-driven array design parameter modeling (ADPM) method.
2) The DCT is performed based on the BEPs to alleviate the computational costs of training the BEPs. Compared with pattern learning based on all-angle information (AAI), this approach leads to significantly reduced training and prediction times while ensuring prediction accuracy.
3) The proposed BEM-based SFMAA synthesis method is adaptable and flexible. The same training samples can be used for different numbers of elements and various pattern problems by only modifying the optimization objectives and fitness function. In addition, LF verification and local optimization are introduced to compensate for the BEM errors and to reduce the number of simulations of the time-consuming high-fidelity (HF) model.

The remainder of the article is organized as follows. Section II presents the BEM method in detail. The efficient SFMAA synthesis method combined with the optimization algorithm is proposed in Section III. Synthesis applications with three types of SFMAAs are utilized to verify the superiority of the proposed SFMAA synthesis method in Section IV. Finally, Section V concludes the article.

II. BE Modeling

In this section, the BE-based SFMAA characterization method is described. Then, the BEPs are trained in the sparse cosine domain, effectively reducing the computational costs. Next, the BEM framework is proposed. A parametric analysis of the training process based on the cosine domain values is performed. Finally, the BEM method is compared with the conventional ADPM approach.

A. SFMAA Characterization

The SFMAA with N elements is shown in Fig. 1 and the schematic of the SFMAA microwave network cascade is shown in Fig. 2. The conventional ADPM approach shown in Fig. 3(a) is a purely data-driven approach, and the large datasets are expensive to acquire. Furthermore, the training and prediction times are nonnegligible.

First, consider the array structure shown in Fig. 1. The array structure includes N elements and $3N + 2$ design variables. Due to the "curse of dimensionality," the number of initial samples increases exponentially as the number of design variables increases; thus, the initial samples are computationally expensive. To address this issue, a "divide-and-conquer" strategy is used. Since the SFMAA elements appear repeatedly, they can be characterized as three types of BEs: a BE at the starting point (BE-s) with index 1, BEs in the middle (BE-m) with indices 2 to $N - 1$, and a BE at the endpoint (BE-e) with index N. The BE-s and BE-m can be viewed as two-port networks, and the BE-e can be viewed as a one-port network. If the end element is connected to a matching load, the BE-e degenerates to a BE-m. An SFMAA with N elements can be regarded as a cascade with 1 BE-s, $N-2$ BE-m, and 1 BE-e, as shown in Fig. 2(a). Compared to the whole SFMAA, the BE-based structure has fewer design parameters and much lower complexity, thus significantly reducing the time to acquire the initial samples.

ML algorithms are adopted to train the S-parameters and patterns of each BE. GP regression (GPR) is a widely used method due to its good adaptability to small sample sizes, nonlinearities, and complex problems [20], [23]. In this work, GPR is used to train surrogate models. Based on the SFMAA design parameters, the S-parameters and radiation patterns of the corresponding BEs are obtained using surrogate models. The S-parameters and radiation patterns of the SFMAA are then characterized by using the S-parameter cascade and array synthesis method. The specific procedure is described in detail in Section II-C.

B. Cosine-Domain BEP Learning

The complex element pattern in the E-plane needs to be learned in SFMAA synthesis problems. GPR methods cannot be used to learn complex values. The real and imaginary parts of the BEP are adopted as the outputs of the GPR method. Suppose there are K angles, and $2K$ surrogate models are trained via traditional AAI training. The inputs to the ML model are the BE structural parameters. The outputs of the ML model are the real part Y_{iR} and the imaginary part Y_{iI}, $i = 1, \ldots, K$, of the BEP, as shown in Fig. 3(c). The number of angles is related to the angle range of interest and the sampling interval. To ensure the accuracy of the pattern information, the angle interval is usually dense. For example, in the pattern between $-180°$ and $179.9°$, consider a point in a $0.1°$ angle interval. Then, each sample has 3600 angles. The training and prediction times for a model trained based on this large amount of training data are not negligible. The training data must be compressed to reduce the computational costs of the training and prediction processes.

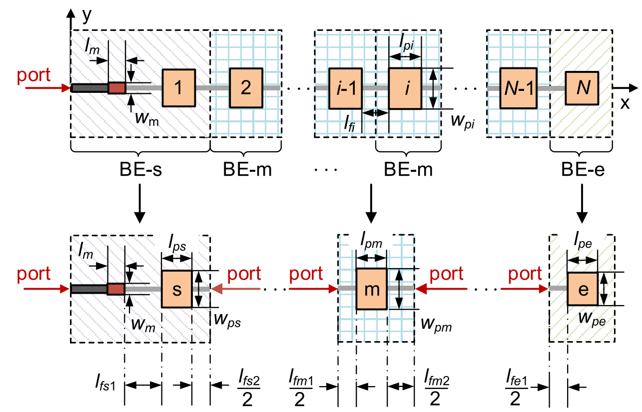


Fig. 1. Schematics of an SFMAA and BEs.

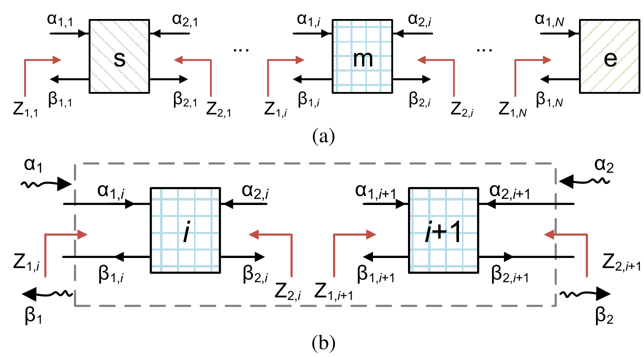


Fig. 2. Schematic of the SFMAA microwave network cascade. (a) SFMAA N-port network cascade. (b) Two-port network cascade.

Inspired by image-processing schemes, many real-world signals are sparse or compressible in certain transform domains. Overcomplete dictionaries are effective in providing sparse signal representations, and the determination of the best sparse representation is NP-hard (nondeterministic polynomial-time hard) [29]. DCT is a fixed dictionary transformation that has been widely used in the fields of image and video processing [30], [31]. The spatial data of an image are mapped to the frequency domain by DCT [31], and the sparse coefficients of the transform domain are used to reduce the input data size. In this study, DCTs are independently

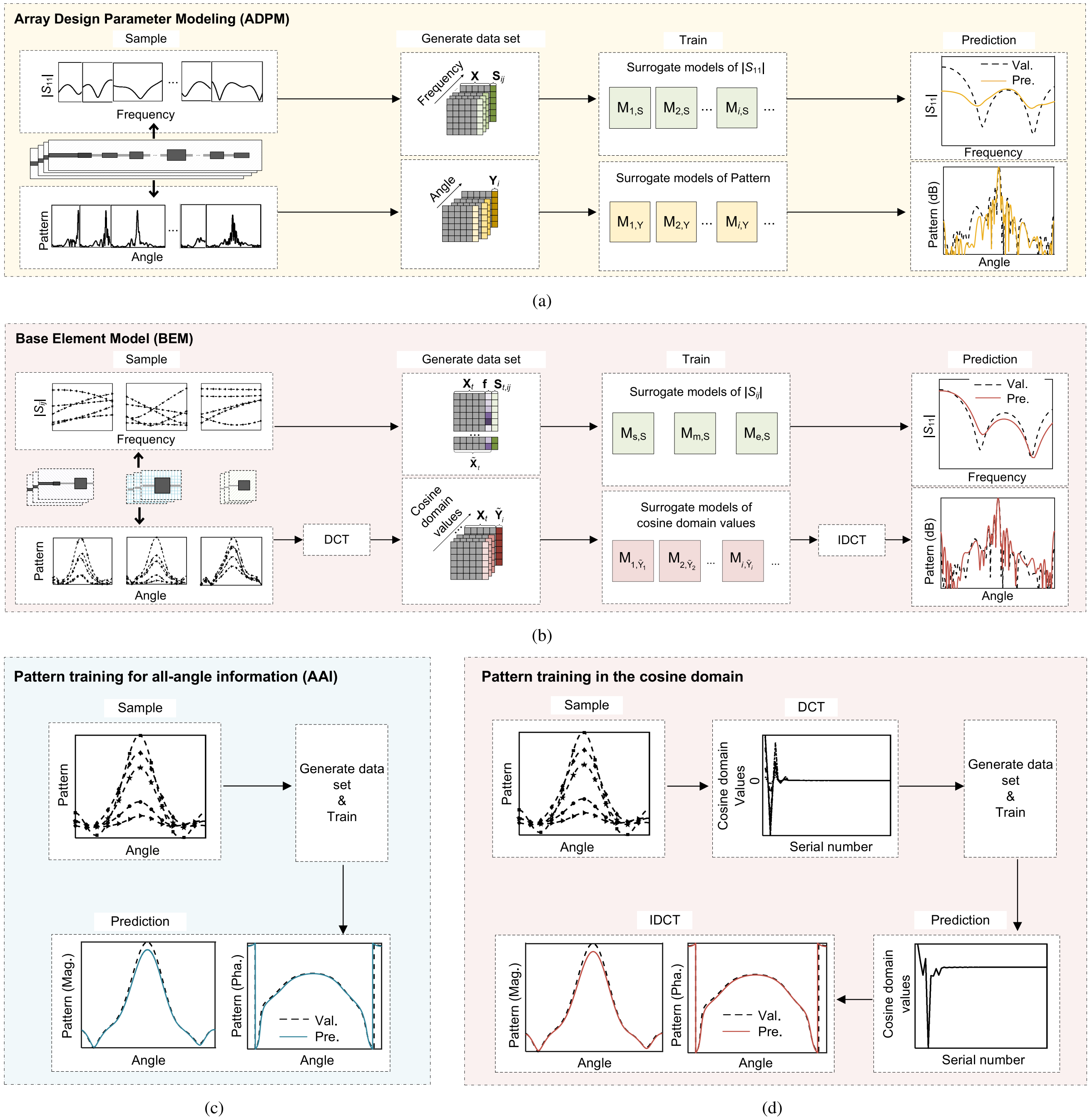


Fig. 3. Flow diagrams of the ADPM, BEM, AAI, and cosine domain learning methods. (a) ADPM. (b) BEM. (c) AAI pattern training. (d) Cosine-domain pattern training.

performed for the real and imaginary parts of the BEPs, denoted as

$$\tilde{\mathbf{Y}} = \mathbf{C}\mathbf{Y} \tag{1}$$

where $\mathbf{Y} \in \mathbb{R}^{K\times 1}$ denotes the real or imaginary part of the BEP, $\tilde{\mathbf{Y}} \in \mathbb{R}^{K\times 1}$ denotes the cosine-domain values, and $\mathbf{C} \in \mathbb{R}^{K\times K}$ denotes the DCT matrix with elements

$$c_{lk} = \begin{cases} \sqrt{\dfrac{1}{K}}, & l = 1,\ 1 \le k \le K \\ \sqrt{\dfrac{2}{K}} \cos \dfrac{\pi(2k-1)(l-1)}{2K}, & 2 \le l \le K,\ 1 \le k \le K. \end{cases} \tag{2}$$

Most elements in $\tilde{\mathbf{Y}}$ are approximately 0, as shown in Fig. 3(d). The cosine domain $\tilde{\mathbf{Y}}$ may not be the sparsest representation of $\mathbf{Y}$, but the fixed transform domain reduces the time required to produce the dictionary. For BEP learning in the cosine domain, the input parameters of the ML model are the structural parameters of the BEs, but the output parameters are the former N_a cosine-domain values $\tilde{Y}_i$ with $i = 1, \ldots, N_a$. In the prediction process, the cosine-domain values are first predicted by the surrogate models and then padded by 0. Finally, the real and imaginary parts of the BEPs are obtained by inverse DCT (IDCT).

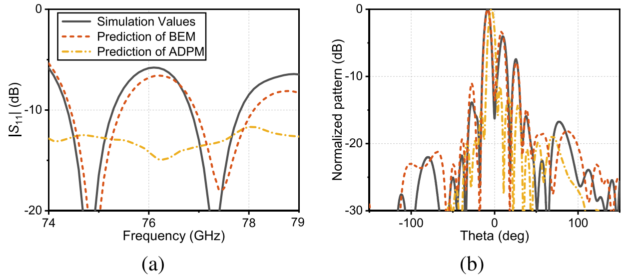


Fig. 4. Typical simulation and prediction curves for BEM and ADPM. (a) $|S_{11}|$. (b) Radiation patterns at 76.5 GHz.

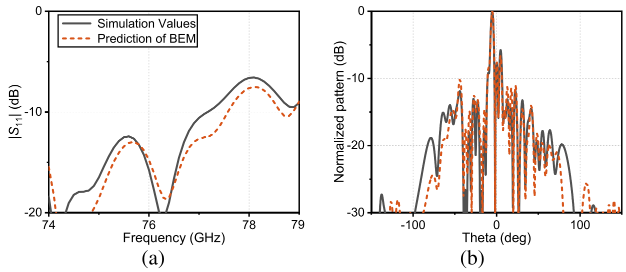


Fig. 5. Typical simulation and prediction curves of different element arrays with $N = 24$ using BEM. (a) $|S_{11}|$. (b) Radiation patterns at 76.5 GHz.

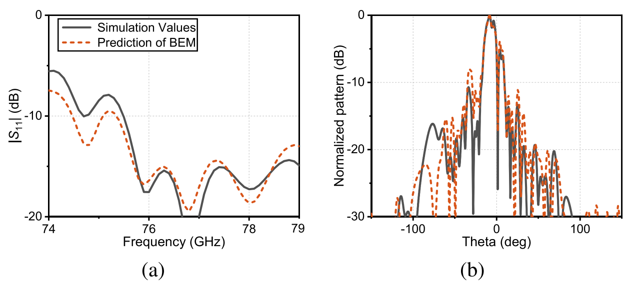


Fig. 6. Typical simulation and prediction curves of different element arrays with $N = 30$ using BEM. (a) $|S_{11}|$. (b) Radiation patterns at 76.5 GHz.

C. BEM Framework

The flow diagram is shown in Fig. 3(b), and the steps in the BEM procedure are described as follows.

Step 1. Initial Sample Generation: First, the SFMAA is represented by three types of BEs, as shown in Fig. 1. The BE-s, BE-m, and BE-e design variables are $[l_m, w_m, l_{ps}, w_{ps}, l_{fs1}, l_{fs2}]$, $[l_{pm}, w_{pm}, l_{fm1}, l_{fm2}]$, and $[l_{pe}, w_{pe}, l_{fe1}]$, respectively. Each BE is sampled individually using Latin hypercube sampling (LHS) [32], and the initial number of samples for each BE is five times the number of design variables, denoted as $N_t = \{30, 20, 15\}, t = \{s, m, e\}$. Next, the initial samples are simulated using full-wave simulations to obtain the S-parameters and radiation patterns.

Step 2. Surrogate Model Training: In this step, GPR is used to train the S-parameters and patterns of each BE. For a p-port network, the real and imaginary parts of S_{ij} must be trained separately, where $i, j = 1, \cdots, p$. Therefore, $2p^2$ surrogate models need to be trained. Due to the reciprocity of the BE-s and BE-m networks, i.e., $\mathbf{S}_{12} = \mathbf{S}_{21}$, a total of 14 surrogate models need to be trained for the S-parameters of the BEM. To predict the S-parameters of a frequency band, the frequency is added to the training set as a feature dimension, and the training set of the S-parameters is expressed as

$$\mathbf{D}_{t,ij} = \left\{ \left(\tilde{\mathbf{x}}_{t,n}, \mathbf{S}_{t,n,ij} \right) | n = 1, \ldots, N_t \right\} \tag{3}$$

where

$$\tilde{\mathbf{x}}_{t,n} = \left[\mathbf{x}_{t,n}, \mathbf{f} \right] \tag{4}$$

where $\mathbf{x}_{t,n}$ is the nth sample of BE-t, and the column vector $\mathbf{f}$ contains the frequency points within the target band. In the training and prediction of the S-parameters, the real and imaginary parts are determined independently.

The BEP is learned in the cosine domain using the method described in Section II-B.

Step 3. Calculation of the S*-Parameters:* After determining the parameter combinations for the arrays, the S-parameter surrogate models obtained in *Step* 2 are used to predict the S-parameters for each BE. Each array can be viewed as a cascade of $N - 1$ two-port networks and a one-port network, as shown in Fig. 2(a). Consider the two-port cascading network in Fig. 2(b) as an example. The right end of the two-port network i is cascaded to the left end of the two-port network $i + 1$, where

$$Z_{2,i} = Z_{1,i+1} \tag{5a}$$
$$\alpha_{2,i} = \beta_{1,i+1} \tag{5b}$$
$$\beta_{2,i} = \alpha_{1,i+1}. \tag{5c}$$

The cascaded S-parameter can be expressed as [33]

$$S_{11} = S_{11,i} + S_{12,i}(1 - S_{11,i+1}S_{22,i})^{-1}S_{11,i+1}S_{21,i} \tag{6a}$$
$$S_{12} = S_{12,i}(1 - S_{11,i+1}S_{22,i})^{-1}S_{12,i+1} \tag{6b}$$
$$S_{21} = S_{21,i+1}(1 - S_{22,i}S_{11,i+1})^{-1}S_{21,i} \tag{6c}$$
$$S_{22} = S_{22,i+1} + S_{21,i+1}(1 - S_{22,i}S_{11,i+1})^{-1}S_{22,i}S_{12,i+1}. \tag{6d}$$

The S-parameters of the array are obtained via recursive operations.

Step 4. Radiation Pattern Calculation: The BEPs are obtained by predicting the real and imaginary parts of the patterns based on the surrogate models obtained in *Step* 2. The excitation of a BE is approximated by calculating the port incident and reflected waves of the one- or two-port network. The current at the reference planes at the two ends of network i can be expressed as [33]

$$I_{1,i} = \frac{1}{\sqrt{Z_{1,i}}}(\alpha_{1,i} - \beta_{1,i}) \tag{7a}$$
$$I_{2,i} = \frac{1}{\sqrt{Z_{2,i}}}(\alpha_{2,i} - \beta_{2,i}). \tag{7b}$$

The positive direction along the X-axis is specified as the reference direction. Because $I_{1,i}$ and $I_{2,i}$ have opposite directions, the total current can be expressed as

$$I_i = I_{1,i} - I_{2,i}. \tag{8}$$

Suppose that the interelement radiative coupling can be ignored; then, the radiation pattern can be written as

$$F(\theta) = \sum_{i=1}^{N} I_i f_i(\theta) \exp\left(\frac{j2\pi \cos(\theta) x_i}{\lambda_0} \right) \tag{9}$$

TABLE I
COMPARISON OF BE-s WITH DIFFERENT N_a AND CONVENTIONAL AAI TRAINING

	N_a			ML model with
	20	30	50	AAI
Mag. MSE	8.54E-03	8.48E-03	8.46E-03	8.76E-03
Pha. MSE	9.16E-03	8.84E-03	8.84E-03	9.43E-03
Train (s)	5.84E-01	8.71E-01	1.52E+00	9.94E+01
Pre. (s)	6.83E-03	1.10E-02	1.47E-02	9.93E-01

TABLE II
COMPARISON OF BE-m WITH DIFFERENT N_a AND CONVENTIONAL AAI TRAINING

	N_a			ML model with
	20	30	50	AAI
Mag. MSE	6.70E-03	6.70E-03	6.70E-03	5.82E-03
Pha. MSE	2.68E-02	2.70E-02	2.70E-02	2.43E-02
Train (s)	5.16E-01	1.08E+00	1.45E+00	8.26E+01
Pre. (s)	1.09E-02	1.13E-02	1.73E-02	9.44E-01

TABLE III
COMPARISON OF BE-e WITH DIFFERENT N_a AND CONVENTIONAL AAI TRAINING

	N_a			ML model with
	20	30	50	AAI
Mag. MSE	1.71E-01	1.70E-01	1.70E-01	2.49E-01
Pha. MSE	5.20E-03	4.57E-03	4.48E-03	4.46E-03
Train (s)	3.92E-01	7.10E-01	1.17E+00	7.46E+01
Pre. (s)	6.29E-03	9.68E-03	1.58E-02	8.37E-01

where I_i is the excitation, $f_i(\theta)$ is the radiation pattern of the ith element, λ_0 is the free space wavelength, and x_i is the absolute position of the BE in the array. The geometric center of the element is used here to approximate the position.

D. Effect of the Number of Cosine Domain Values

The sizes of the training sets are 30, 20, and 15. The test set for each BE includes ten samples. Consider the pattern from $-180°$ to $179.9°$ and take a point in the angle interval of $0.1°$. There are 3600 angles. Tables I to III show the mean square errors (mses) of the magnitude and phase, the training time, and the prediction time for cosine domain training and AAI training. Due to the periodicity of the phase, the phase MSE is calculated based on the cosine of the phase. N_a are 20, 30, and 50. The prediction accuracy does not improve significantly with increasing N_a. To balance the training time, prediction time, and precision, N_a is set to 20 in this work. Compared with the training time of the traditional AAI training method, the training time of the proposed method is reduced from tens of seconds (s) to approximately 1 s, and the prediction time is reduced from approximately 1 s to less than 0.01 s while maintaining similar prediction accuracy.

E. Comparison Between BEM and ADPM

The BEM and ADPM methods are compared in this section. A 12-element SFMAA operating at 77 GHz is presented as an example. There are 38 design variables, including the length and width of the quarter-wave transformer and elements and the interelement spacing, denoted as $[l_m, w_m, l_{pi}, w_{pi}, l_{fi}]$, where $i = 1, \ldots, 12$. The lower bounds are [0.50, 0.30, 1.04, 0.13, 1.00], and the upper bounds are [0.65, 0.50, 1.14, 1.62, 1.40]. The simulation time of CST Microwave Studio [34] is approximately 1960 s. The training set N_w for the ADPM method is 190. The optimization ranges of l_{ps}, l_{pm}, and l_{pe} are equal to l_{pi}; w_{ps}, w_{pm}, and w_{pe} are equal to w_{pi}; and l_{fs1}, l_{fs2}, l_{fm1}, l_{fm2}, and l_{fe1} are equal to l_{fi}. The simulation times of the BE-s, BE-m, and BE-e models are 392, 460, and 252 s, respectively, and the respective training sets include 30, 20, and 15 elements, respectively. The sampling times of the ADPM and BEM methods are 103 and 6.87 h, respectively.

TABLE IV
TRAINING TIME, PREDICTION TIME, AND MMSE OF THE BEM AND ADPM METHODS

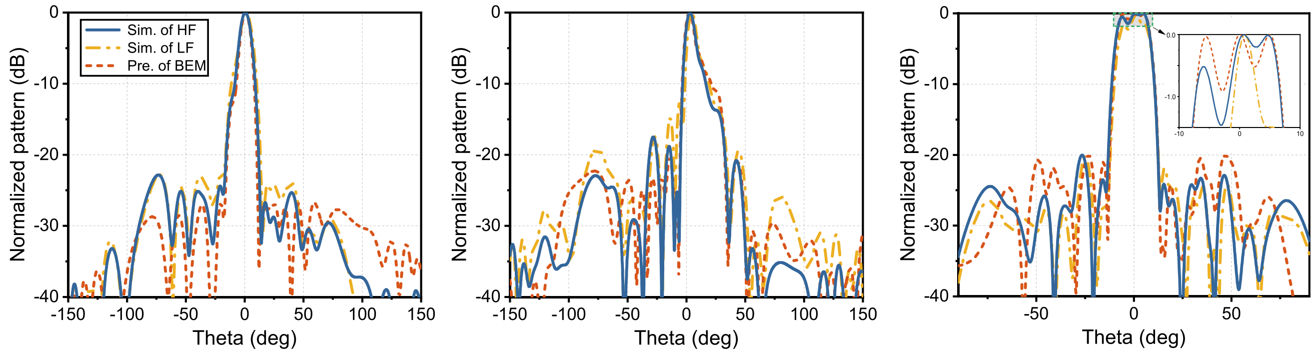
		Train (s)	Pre. (s)	MMSE
BEM	S-parameter	3.81E+01	6.15E-02	8.97E-03
	Radiation pattern	5.53E+00	1.48E-01	4.27E-03
ADPM	S-parameter	1.03E+01	2.47E-02	6.73E-02
	Radiation pattern	1.04E+03	2.18E+00	1.10E-02

The comparison results of the BEM and ADPM methods are shown in Table IV. The training set is S_{11} at $74-79$ GHz and the radiation pattern at 76.5 GHz. Typical simulation and prediction curves for BEM and ADPM are shown in Fig. 4. The design parameter combinations of SFMAA with N $=$ 12 are given randomly. Notably, the training dimension of ADPM is high in this case, and the prediction results when the frequency dimension is included are worse than those without the frequency dimension. The S-parameters of the ADPM method are trained separately for the real and imaginary parts of the S-parameters for 51 frequency points in the $74-79$ GHz range. The training and prediction times of the BEM method are shorter than those of the ADPM method, except for the training time of the S-parameter, and the prediction accuracy of the BEM method is higher than that of the ADPM method. The radiation pattern synthesis with the BEM method requires the S-parameters of the BEs. Therefore, the training time for the radiation pattern includes the training time for some S-parameters, and the model learns the S-parameters of the BEs at 75.9–77.1 GHz for 13 frequency points.

In addition, the prediction with the BEM method does not limit the number of elements in the array, while changing the number of elements in the array requires resampling with the ADPM method. The design parameter combinations of SFMAA for $N = 24$ and $N = 30$ are given randomly, and the corresponding simulation results and BEM predictions are shown in Figs. 5 and 6.

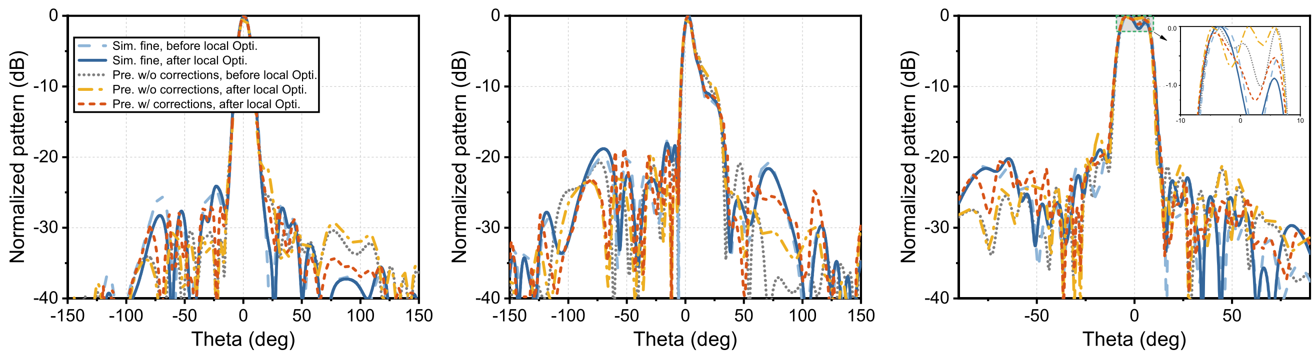
III. MLA-SFMAA SYNTHESIS

In this section, the MLA-SFMAA synthesis based on the BEM method combined with an optimization algorithm is proposed. The steps in the MLA-SFMAA scheme are described as follows.



(a) (b) (c)

Fig. 7. Comparison of radiation patterns at 76.5 GHz for HF simulation, LF simulation, and BEM prediction. (a) $N = 12$, low SLL synthesis. (b) $N = 14$, cosecant squared radiation pattern synthesis. (c) $N = 24$, flat-top radiation pattern synthesis.



(a) (b) (c)

Fig. 8. Comparison of radiation patterns at 76.5 GHz for HF simulation results and prediction curves with and without corrections. (a) $N = 12$, low SLL synthesis. (b) $N = 14$, cosecant squared radiation pattern synthesis. (c) $N = 24$, flat-top radiation pattern synthesis.

Step 1. Initial Settings: First, the number of elements in the array N, the design variables of the arrays and BEs l_m, w_m, l_{pi}, w_{pi}, l_{fsi}, l_{ps}, w_{ps}, l_{fs1}, l_{fs2}, l_{ms}, w_{ms}, l_{fm1}, l_{fm2}, l_{es}, w_{es}, and l_{es1}, the design goals and constraints, and the number of data points in the training sets N_s, N_m, and N_e are defined. Then, LHS and the full-wave simulations of the BEs are implemented to generate the training sets.

Step 2. Train the Surrogate Models: The method described in *Step* 2 of Section II is used to construct surrogate models for the *S*-parameters and radiation patterns of the BEs. The MATLAB *fitrgp* function is used to train the surrogate models.

Step 3. Optimization and Validation: A global optimization algorithm is first implemented to search for the optimal parameter combination based on the surrogate models. The prediction process is the method described in *Steps* 3 and 4 of Section II. The search engine is GA and *GA* in MATLAB is used.

For high-dimensional optimization problems, global searches are likely to fall into local optima. To get rid of the local optima, most methods adopt multiple restarts [35]. If the optimal value obtained by the algorithm is worse than the preset threshold, the optimization process is restarted until the threshold is reached or the maximum number of restarts I_{GA} is reached. The multiple restart optimization approach in this study increases the likelihood of synthesizing to the target design but does not guarantee optimization to the global optimum. In addition, the coupling between elements in arrays is ignored during BEM, leading to errors in the synthesized array patterns. To reduce the simulation times of the expensive HF model, the LF model is first used for verification.

A comparison of radiation patterns at 76.5 GHz for the HF simulation, LF simulation, and BEM prediction is shown in Fig. 7. The simulation curves of both HF and LF are obtained from CST simulation, where Fig. 7(a) shows an example of low SLL synthesis, the number of meshes of the HF model is about 5.6 million, and the simulation time is about 1960 s. The LF model is similar to the HF model but with a number of meshes of 0.3 million, and the simulation time is about 104 s. Fig. 7(b) is the synthesized cosine-squared radiation pattern, the number of meshes of the HF model is about 6.7 million, and the simulation time is about 2185 s; the number of meshes of the LF model is 0.5 million, and the simulation time is about 140 s. Fig. 7(c) presents the synthesized flat-top radiation pattern, the HF model has about 12 million meshes and the simulation time is about 6032 s. The LF model has 0.5 million meshes and the simulation time is about 132 s.

In the SLL region, especially the low SLL synthesis region in Fig. 7(a), the SLL value predicted by the BEM method is lower than the HF simulation value, while the SLL value of the LF method is more consistent with the HF result. However, the LF result has a larger deviation than the BEM result in the mainlobe region. The LF model can capture the features of the radiation patterns but it is not particularly good at representing the reflection response [16], [36]. Based on the simulation results shown in Fig. 7, the LF simulation is carried out after the optimization process. The predicted BEM values are used for the mainlobe and S-parameters, while the simulated LF values are used for the SLL. The optimization and LF validation process is repeated I_{LF} times or stops when the threshold value is reached. The optimal design parameters $\mathbf{x}_{\mathrm{b}}$ are verified by HF simulations, and the S-parameter $R_s(\mathbf{x}_{\mathrm{b}})$ and the radiation pattern $R_p(\mathbf{x}_{\mathrm{b}})$ are recorded.

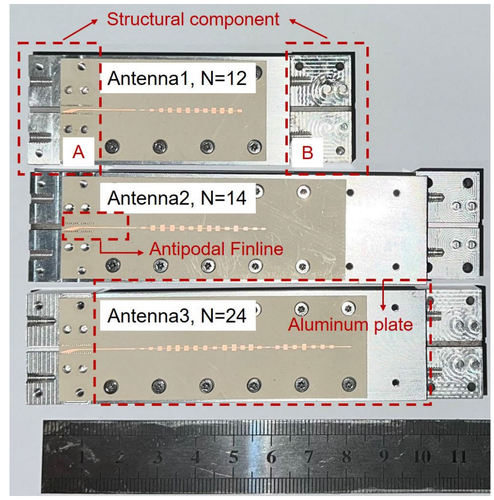
Step 4. Local Optimization: If the HF simulation results are close to the optimization objectives and constraints but do not satisfy the stop conditions, local optimization is performed in the neighborhood of $\mathbf{x}_{\mathrm{b}}$, that is, the parameters that are most sensitive to unsatisfied objectives or constraints are optimized.

For example, in some cases, the SLL or half-power beamwidth (HPBW) results may not meet the design specifications. The magnitude of the element excitation may greatly impact these specifications, and the $\mathbf{x}_{\mathrm{b}}$ parameters are fixed except w_{pi}. In some cases, the main direction angle (denoted as α) does not match the targets, the phase of the element excitation and the interelement spacing may affect it; in this case, l_{fi} and l_{pi} are locally optimized. If the S-parameters are not consistent with the goal, the parameters are first parametrically scanned using surrogate models for each parameter, and the N_{SA} parameter, which is sensitive to the S-parameters and not sensitive to the radiation pattern, is locally optimized.

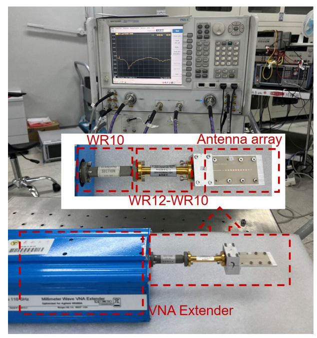
Moreover, local optimization is conducted in the neighborhood of $\mathbf{x}_{\mathrm{b}}$ in *Step* 3. The simulated and predicted values of the $\mathbf{x}_{\mathrm{b}}$ radiation pattern are expressed as $R_p(\mathbf{x}_{\mathrm{b}})$ and $U_p(\mathbf{x}_{\mathrm{b}})$, respectively. The residuals between the simulated and predicted values in the neighborhood of $\mathbf{x}_{\mathrm{b}}$ are assumed to be $R_p(\mathbf{x}_{\mathrm{b}}) - U_p(\mathbf{x}_{\mathrm{b}})$. In this case, the radiation prediction and the residual are used to correct the local optimization results. The residuals at each angle are limited to the range $\pm r$ to prevent excessive differences at some angles.

A typical comparison of radiation patterns at 76.5 GHz for the HF simulation and prediction curves with and without this correction is shown in Fig. 8. Taking Fig. 8(b) as an example, the width of the SFMAA element is locally optimized when all the design specifications are satisfied except for the SLL before local optimization. The correction value in the local optimization is $R_p(\mathbf{x}_{\mathrm{b}}) - U_p(\mathbf{x}_{\mathrm{b}})$, and comparing the red dashed line and the yellow dash-dotted line, it is possible to get a more accurate prediction after correction, especially in the mainlobe area. Although this hypothesis is not sufficiently rigorous, the possibility of finding a combination of parameters that satisfies the optimal objective can be improved by the local optimization process. In the fitness function of the local optimization process, the weight of the target to be optimized should be increased.

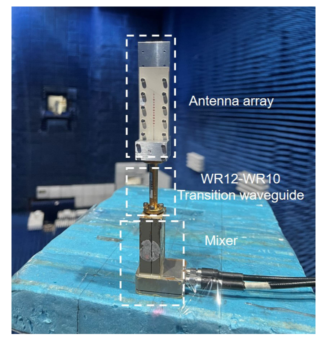
Step 5. Termination Conditions: The optimization process is stopped if the HF full-wave simulation results meet the design specifications or the optimization process reaches the maximum number of iterations. Otherwise, the region near the current HF array simulation value is considered a forbidden area for optimization. The optimization process then continues, and the model returns to *Step* 3.



(a)



(b)



(c)

Fig. 9. Photographs of the fabricated prototypes for three types of antenna arrays. (a) Antenna array prototypes. (b) Measurement setup for the S-parameters. (c) Measurement setup for the radiation patterns.

TABLE V
COMPUTATION TIMES FOR THE THREE EXAMPLES

Num.	Opti. (h)	HF (h)	LF (h)	Overall (h)
Antenna 1	0.11	1.09	0.06	8.13
Antenna 2	5.29	13.96	2.92	29.04
Antenna 3	8.13	33.51	2.82	51.33

IV. VERIFICATION EXAMPLES

In this section, three practical examples, low SLL radiation pattern synthesis, cosecant-squared radiation pattern synthesis, and flat-top radiation pattern synthesis, are simulated and fabricated to verify the performance of the proposed MLA-SFMAA synthesis method. The prototypes and measurement setup of the antenna array are shown in Fig. 9.

A. Low SLL Array Synthesis

The HPBW reference value in the elevation plane for long-range radar is 10° [37], and the number of elements is selected to be 12 [9]. A low SLL is required for the array to reduce the influence of ground clutter on the radar system and improve radar performance. This example requires that the SLL value at 76.5 GHz be as small as possible when the HPBW is less than 10°, the α is within $\pm 1°$, and $|S_{11}|$ within $76 - 77$ GHz is less than -10 dB. The stop condition is that the SLL value is less than -25 dB or the maximum

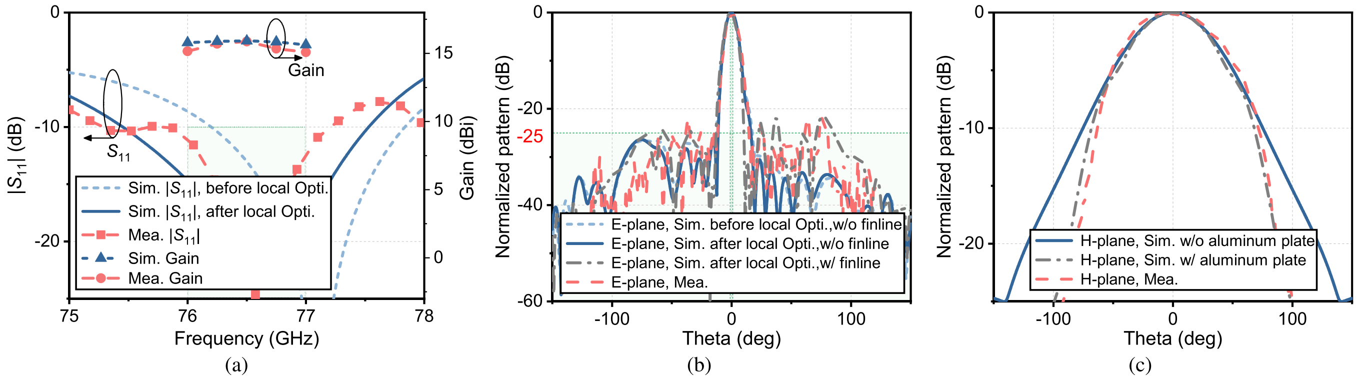


Fig. 10. Simulated and measured results of low-SLL SFMAA synthesis with $N = 12$. (a) $|S_{11}|$ and gain. (b) E-plane pattern at 76.5 GHz. (c) H-plane pattern at 76.5 GHz.

number of iterations is 20. The sampling setting of the BEs and the training process of surrogate models are the same as those described in Section II-E. However, in this case, the S-parameter model determines the response value in the frequency band of 75.9–77.1 GHz with a frequency interval of 0.1 GHz. The fitness function in the optimization process can be described as

$$f(\mathbf{x}_i) = \min_{\mathbf{x}_i}(\tilde{U}_{\text{SLL}} + 2^{\hat{U}_{S_{11}}} + 2^{\hat{U}_{\alpha}} + 5^{\hat{U}_{\text{HPBW}}}) \tag{10}$$

where

$$\hat{U}_{S_{11}} = \max\big(0, 10\max\big(\tilde{U}_{S_{11}} - 0.3548\big)\big) \tag{11a}$$

$$\hat{U}_{\alpha} = \max\big(0, |\tilde{U}_{\alpha}| - 1\big) \tag{11b}$$

$$\hat{U}_{\text{HPBW}} = \max\big(0, \tilde{U}_{\text{HPBW}} - 10\big). \tag{11c}$$

$\tilde{U}_{\text{SLL}}$, $\tilde{U}_{\alpha}$, and $\tilde{U}_{\text{HPBW}}$ are the SLL, α, and HPBW calculated based on the BEM predictions and the synthesized radiation pattern. $\tilde{U}_{S_{11}}$ is obtained by cascading the $|S_{11}|$ values predicted by the BEM method. $|S_{11}|$, HPBW, and α are added as exponential penalties to the fitness function to serve as model constraints. In addition, the constraint on HPBW is tighter in this example, so the exponential penalty with base 5 is chosen. Although the target of $|S_{11}|$ is -10 dB, this target is relaxed to -9 dB (0.3548) in the optimization process due to the prediction error. The simulation curves of the LF and HF models are shown in Fig. 7(a). The optimized S_{11} and the radiation patterns at 76.5 GHz are shown in Fig. 10. The optimization time is shown in Table V, where the sampling and training times are 6.87 h and 5.53 s, respectively.

The blue dashed line in Fig. 10 satisfies the objectives and constraints except for $|S_{11}|$. Thus, local optimization is performed. In this case, $N_{\text{SA}} = 12$. The local variation range of N_{SA} is $x_b \pm \delta \times 0.1$, where δ is the parameter bound described in Section II-E. The fitness function for the local optimization process is given by

$$f(\mathbf{x}_i) = \min_{\mathbf{x}_i}\Big(\max(\tilde{U}_{S_{11}}) + 10^{\hat{U}_{\text{SLL}}} + 10^{\hat{U}_{\alpha}} + 10^{\hat{U}_{\text{HPBW}}}\Big) \tag{12}$$

where

$$\hat{U}_{\text{SLL}} = \max\Big(0, \widehat{U}_{\text{SLL}} + 25\Big) \tag{13a}$$

$$\hat{U}_{\alpha} = \max\Big(0, |\widehat{U}_{\alpha}| - 1\Big) \tag{13b}$$

$$\hat{U}_{\text{HPBW}} = \max\Big(0, \widehat{U}_{\text{HPBW}} - 10\Big) \tag{13c}$$

and $\widehat{U}_{(\cdot)}$ is the BEM synthesis radiation pattern with the added correction term. After the local optimization scheme is performed, the $|S_{11}|$ value within the frequency band is less than -10 dB, $\alpha = -0.3°$, and the HPBW is 9.3°. The prototype antenna is shown in Fig. 9(a), and the measurement setups for the S-parameters and radiation patterns are shown in Fig. 9(b) and (c), respectively. The antipodal finline transition model is used for the waveguide-to-microstrip transition [38]. The structural components A and B in Fig. 9(a) are assembled to connect the transition waveguide. The measurement results are shown in Fig. 10. The effects of the transition waveguide and the antipodal finline transition model are determined based on the measured S-parameters, and the loss of the antipodal finline transition model is compensated in the gain measurement. The measured SLL is -21.89 dB, and the HPBW is 9.5° due to the influence of the structural components and the antipodal finline transition model. In addition, the aluminum plate used for support narrows the H-plane radiation pattern, as shown in Fig. 10(c).

B. Cosecant-Squared Radiation Pattern Synthesis

The cosecant-squared pattern ensures identical power coverage at different distances from the antenna, which meets certain scenario requirements [39]. The number of array elements is chosen to be 14. The cosecant squared pattern formation region is defined as 0°–30°, and the target pattern can be expressed as

$$f(\theta) = \begin{cases} 20\log_{10}\left(\dfrac{\csc(\theta + 5°)}{\csc(8°)}\right), & \theta \in [3°, 30°] \\ 0, & \theta \in [0°, 3°] \end{cases} \tag{14}$$

and $|S_{11}|$ within 76–77 GHz is less than -10 dB. The SLL defined in $[-150°, -5°] \cup [35°, 150°]$ is less than -20 dB. Moreover, there is no SLL in the transition area of $[-5°, 0°] \cup [30°, 35°]$. The fitness function in the optimization process is

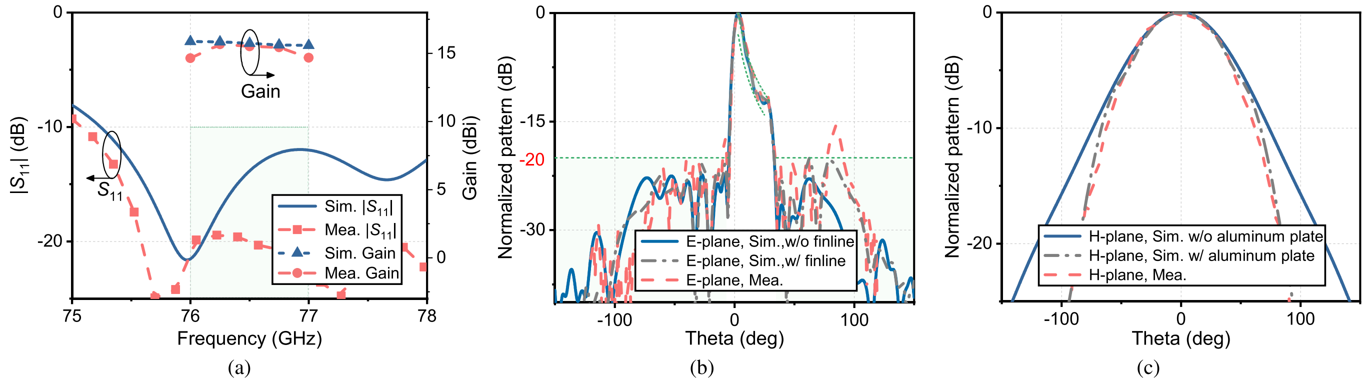


Fig. 11. Simulated and measured results of cosecant-squared radiation pattern synthesis with $N = 14$. (a) $|S_{11}|$ and gain. (b) E-plane pattern at 76.5 GHz. (c) H-plane pattern at 76.5 GHz.

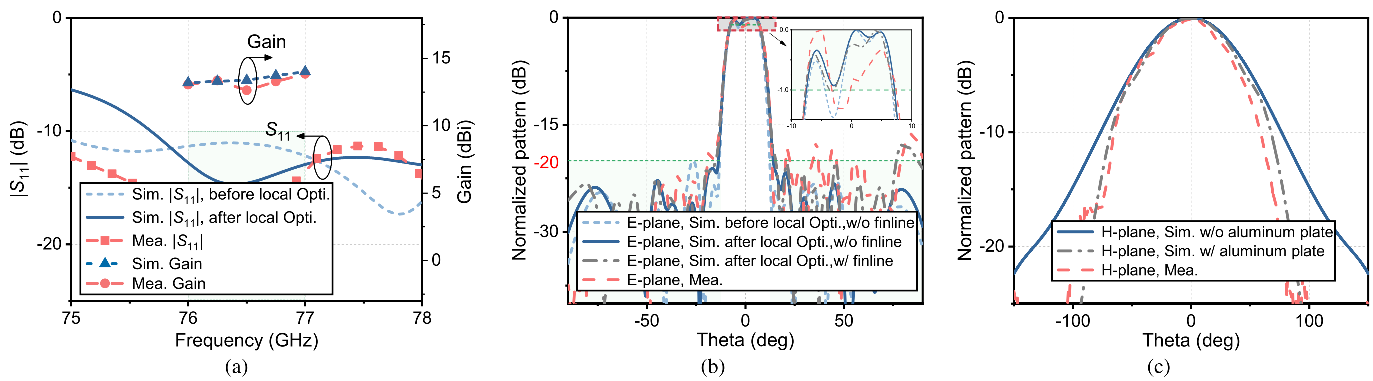


Fig. 12. Simulated and measured results of flat-top radiation pattern synthesis with $N = 24$. (a) $|S_{11}|$ and gain. (b) E-plane pattern at 76.5 GHz. (c) H-plane pattern at 76.5 GHz.

expressed as

$$f(\mathbf{x}_i) = \min_{\mathbf{x}_i}\left(\sum_{m=1}^{3} w_m \hat{f}_{\theta_m}(\mathbf{x}_i) + \sum_{n=4}^{5} 5^{\hat{f}_{\theta_n}(\mathbf{x}_i)} + 2^{\hat{U}_{S_{11}}}\right) \quad (15)$$

and

$$\hat{f}_{\theta_j}(\mathbf{x}_i) = \begin{cases} \sum\left(\max\left(0, \tilde{f}_{\theta_j}(\mathbf{x}_i) + 20\right)\right), & j = 1, 3 \\ \sum\left(\left|\tilde{f}_{\theta_j}(\mathbf{x}_i) - f(\theta_j)\right|\right), & j = 2 \\ \sum\left(\left|\tilde{U}_{\mathrm{SLL}_{\theta_j}}\right|\right), & j = 4, 5 \end{cases} \quad (16)$$

where $\theta_1 \in [-150^\circ, -5^\circ]$, $\theta_2 \in [3^\circ, 30^\circ]$, $\theta_3 \in [35^\circ, 150^\circ]$, $\theta_4 \in [-5^\circ, 0^\circ]$, and $\theta_5 \in [30^\circ, 35^\circ]$. The weight coefficients $\mathbf{w}$ are $w_1 = 0.3$, $w_2 = 0.6$, and $w_3 = 0.1$, and $\tilde{f}_{\theta_j}$ is the synthesis radiation pattern generated by the BEM method in the θ_j region.

The optimization target is relaxed at the discontinuity in the directional pattern. The optimization stopping condition is set to satisfy the SLL and S-parameter constraints, while the difference between the synthesis pattern and the target pattern in 3°–25° less than 3 dB or the maximum number of iterations (20) is reached. The optimized $|S_{11}|$ and the radiation patterns at 76.5 GHz are shown in Fig. 11. The optimization time is shown in Table V. In the sampling and training processes, the optimization objectives are satisfied in 29.04 h. The measurement settings are the same as those described in Section IV-A and are thus not repeated here. The measured and simulated radiation patterns in the cosecant-squared region are shown in Fig. 11(d) and are in good agreement. In the SLL region, the simulation value is −22.36 dB, and the measured value is −15.62 dB.

C. Flat-Top Radiation Pattern Synthesis

The MLA-SFMAA method synthesizes a flat-top radiation pattern with a 24-element array. The optimization problem is described as follows: $[-7.5^\circ, 7.5^\circ]$ is the flat-top region where the local minimum is larger than −1 dB. The SLLs in the region $[-150^\circ, -12^\circ] \cup [12^\circ, 150^\circ]$ are less than −20 dB while satisfying the constraint that $|S_{11}|$ is less than −10 dB in the 76–77 GHz range. The fitness function in the optimization process is the same as (15), where

$$\hat{f}_{\theta_j}(\mathbf{x}_i) = \begin{cases} \sum\left(\max\left(0, \tilde{f}_{\theta_j}(\mathbf{x}_i) + 20\right)\right), & j = 1, 3 \\ \sum\left(\left|\min\left(0, \tilde{f}_{\theta_j}(\mathbf{x}_i) + 1\right)\right|\right), & j = 2 \\ \sum\left(\max\left(\tilde{U}_{\mathrm{SLL}_{\theta_j}} + 20\right)\right), & j = 4, 5 \end{cases} \quad (17)$$

where $\theta_1 \in [-150^\circ, -12^\circ]$, $\theta_2 \in [-7.5^\circ, 7.5^\circ]$, $\theta_3 \in [12^\circ, 150^\circ]$, $\theta_4 \in [-12^\circ, -10^\circ]$, and $\theta_5 \in [10^\circ, 12^\circ]$. $[-10^\circ, -7.5^\circ] \cup [7.5^\circ, 10^\circ]$ are the transition regions. The weight coefficients $\mathbf{w}$ are $w_1 = 0.1$, $w_2 = 0.8$, and $w_3 = 0.1$.

The optimization times are shown in Table V. Before the local optimization process is performed, the local minimum in the flat-top region is -1.46 dB. The parameters adjusted in the local optimization process are the geometric size of each element, $[l_{pi}, w_{pi}]$, and the interelement spacing l_{fi}. The fitness function of the local optimization process is

$$f(\mathbf{x}_i) = \min_{\mathbf{x}_i}\left(\hat{f}_{\theta_2}(\mathbf{x}_i) + \sum_{n=1,n\neq 2}^{5} 5^{\hat{f}_{\theta_n}(\mathbf{x}_i)} + 2^{\hat{U}_{S_{11}}}\right) \quad (18)$$

where

$$\hat{f}_{\theta_j}(\mathbf{x}_i) = \begin{cases} \sum\Big(\max\Big(\widehat{U}_{\mathrm{SLL}_{\theta_j}} + 20\Big)\Big), & j = 1, 3, 4, 5 \\ \sum\Big(\Big|\min\Big(0,\, \widehat{f}_{\theta_j}(\mathbf{x}_i) + 1\Big)\Big|\Big), & j = 2 \end{cases} \quad (19)$$

where $\widehat{f}_{\theta_j}(\mathbf{x}_i)$ is the BEM synthesis radiation pattern with the correction term added, as illustrated in *Step* 4 in Section III.

After the local optimization process is carried out, the local minimum value in the flat-top region is -0.93 dB, and the SLL is -21.14 dB. The optimized $|S_{11}|$ and the radiation patterns at 76.5 GHz are shown in Fig. 12. The measured local minimum value in the flat-top region is -1.32 dB, and the SLL is -16.25 dB.

V. Conclusion and Future Work

An efficient and flexible knowledge-guided BEM method has been proposed for the MLA-SFMAA synthesis problem based on network cascades, array synthesis, and cosine-domain training. The elements in the SFMAA can be characterized by three types of BEs. Each BE is sampled and trained independently, which dramatically reduces the computational costs due to the "curse of dimensionality." To alleviate the BEP training costs, DCTs are implemented based on the BEPs. Compared with conventional AAI training methods, the training and prediction times for learning in the transform domain are significantly reduced, while the prediction accuracy is comparable. The SFMAA responses are obtained by the BEM method using network cascades and array synthesis, which significantly reduces the complexity of the SFMAA synthesis problem. Three types of practical SFMAA synthesis applications, including low SLL synthesis, cosecant squared pattern synthesis, and flat-top synthesis, have been employed to verify the effectiveness of the proposed method. In addition, the designed arrays have been fabricated, and the measured results were consistent with the simulation results.

It is worth noting that the radiation pattern of interest in this work is the copolarization pattern for a single-frequency point. It is also feasible to compress and train the cross-polarized pattern of the BE in the cosine domain. Furthermore, for a broadband pattern synthesis problem, it is necessary to train surrogate models for the pattern at multiple frequency points of interest, and the training time and the prediction time will increase because a new dimension is added. In future work, the proposed BEM method will be extended to solve synthesis problems of broadband dual-polarized series-fed antennas and arrays.

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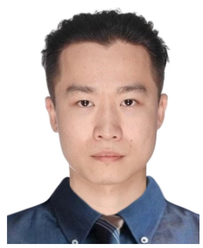
Weiqi Chen (Graduate Student Member, IEEE) was born in 1995. She received the B.S. degree in electronic information science and technology from Nankai University, Tianjin, China, in 2018. She is currently pursuing the Ph.D. degree in electrical engineering with Southeast University, Nanjing, China.

Her current research interests are artificial intelligence (AI)-powered antenna and radio-frequency technologies (iART).



Qi Wu (Member, IEEE) was born in 1993. He received the B.S. degree in communication engineering from the Communication University of China, Beijing, China, in 2012, and the M.S. and Ph.D. degrees in electrical engineering from Southeast University, Nanjing, China, in 2015 and 2019, respectively.

From October 2016 to September 2017, he was a Junior Visiting Fellow at the Ando and Hirokawa Laboratory, Tokyo Institute of Technology, Tokyo, Japan. He joined the State Key Laboratory of Millimeter Waves, Southeast University, in March 2019. He is currently an Associate Professor. He has authored and coauthored over 50 technical publications in peer-reviewed international journals and conference proceedings and over 20 patents. His current research interest is artificial intelligence (AI)-powered antenna and radio-frequency technologies (iART).

Dr. Wu was a recipient of the Honorable Mention at the Student Contest of IEEE APS-URSI Conference in 2018, selected as the Finalist in the IEEE EuCAP in 2016, the student paper contest of iWAT in 2018, and the best poster contest of iWAT in 2019. He has served as a reviewer for the IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION, the IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS, and the IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS.

Jiahao Wei was born in 1998. He received the B.S. degree in electronic science and technology from the Huazhong University of Science and Technology, Wuhan, China, in 2020, and the M.S. degree in electrical engineering from Southeast University, Nanjing, China, in 2023.

Since July 2023, he has been an Engineer with Hisilicon Technologies Company Ltd., Shenzhen, China. His current research interests include machine-learning-assisted ON-chip passive device design.

Chen Yu (Member, IEEE) was born in 1979. She received the M.S. and Ph.D. degrees in electrical engineering from Southeast University, Nanjing, China, in 2004 and 2011, respectively.

She joined the State Key Laboratory of Millimeter Waves, Southeast University, in April 2004. In 2007, she was a Short-Term Visiting Scholar at the Kalmar Institute of Technology, Kalmar, Sweden. She is currently an Associate Professor. Her current research interests include antennas and antenna arrays for wireless mobile communications and radars.

Dr. Yu received the First-Class Science and Technology Progress Award of Jiangsu Province of China in 2009.

Haiming Wang (Member, IEEE) was born in 1975. He received the B.S., M.S., and Ph.D. degrees in electrical engineering from Southeast University, Nanjing, China, in 1999, 2002, and 2009, respectively.

He joined the School of Information Science and Engineering, Southeast University, in 2002. He is currently a Distinguished Professor. He is also a part-time Professor with Purple Mountain Laboratories, Nanjing. He has authored and coauthored over 50 technical publications in IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION and other peer-reviewed academic journals, more than 70 patents, and 52 patents have been granted. His current research interests include artificial intelligence (AI)-powered antenna and radio-frequency technologies (iART), AI-powered channel measurement and modeling technologies (iCHAMM), and integrated communications, sensing, and positioning (iCSAP).

Prof. Wang was awarded twice for contributing to the development of IEEE 802.11aj by the IEEE Standards Association in 2018 and 2020.

Wei Hong (Fellow, IEEE) received the B.S. degree from the University of Information Engineering, Zhengzhou, China, in 1982, and the M.S. and Ph.D. degrees from Southeast University, Nanjing, China, in 1985 and 1988, respectively, all in radio engineering.

In 1993 and from 1995 to 1998, he was a Short-Term Visiting Scholar with the University of California at Berkeley, Berkeley, CA, USA, and the University of California at Santa Cruz, Santa Cruz, CA, USA. He is currently a Professor with the School of Information Science and Engineering, Southeast University. He has authored and coauthored over 400 technical publications and two books. He has been engaged in numerical methods for electromagnetic problems, millimeter-wave theory and technology, antennas, and RF technology for wireless communications.

Dr. Hong is a fellow of CIE and the Vice President of the CIE Microwave Society and Antenna Society. He was an Elected IEEE MTT-S AdCom Member from 2014 to 2016. He was twice awarded the National Natural Prizes and four times awarded the First-Class Science and Technology Progress Prizes issued by the Ministry of Education of China and Jiangsu Province Government. Besides, he also received the Foundations for China Distinguished Young Investigators and for "Innovation Group" issued by the National Natural Science Foundation of China. He has served as an Associate Editor for IEEE TRANSACTION ON MTT from 2007 to 2010.